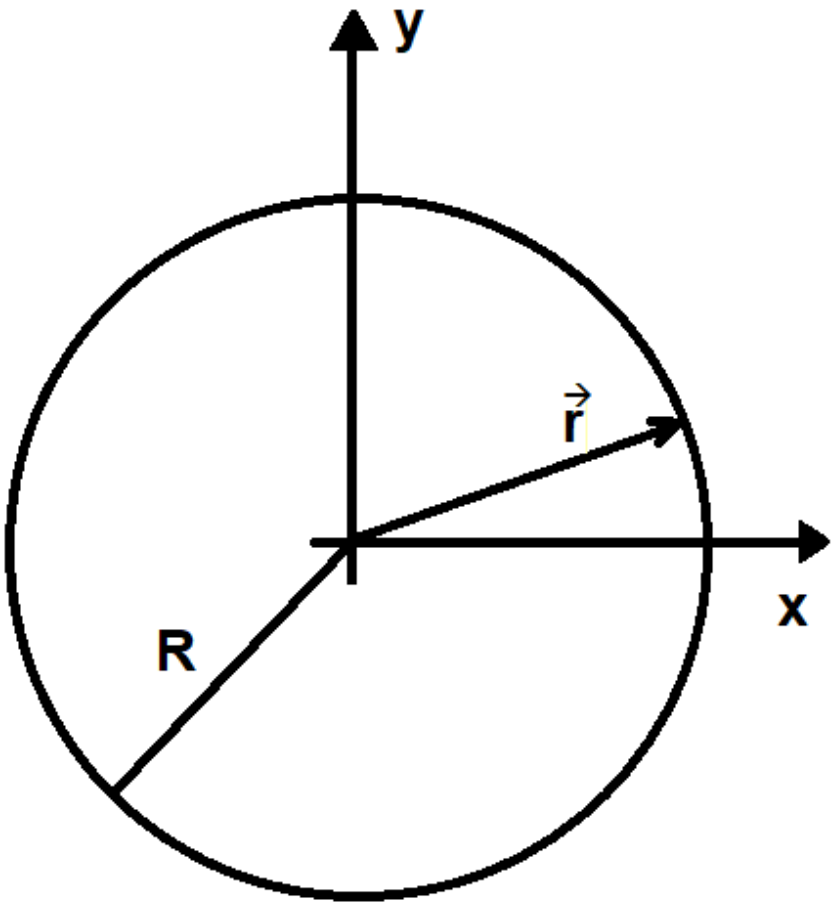
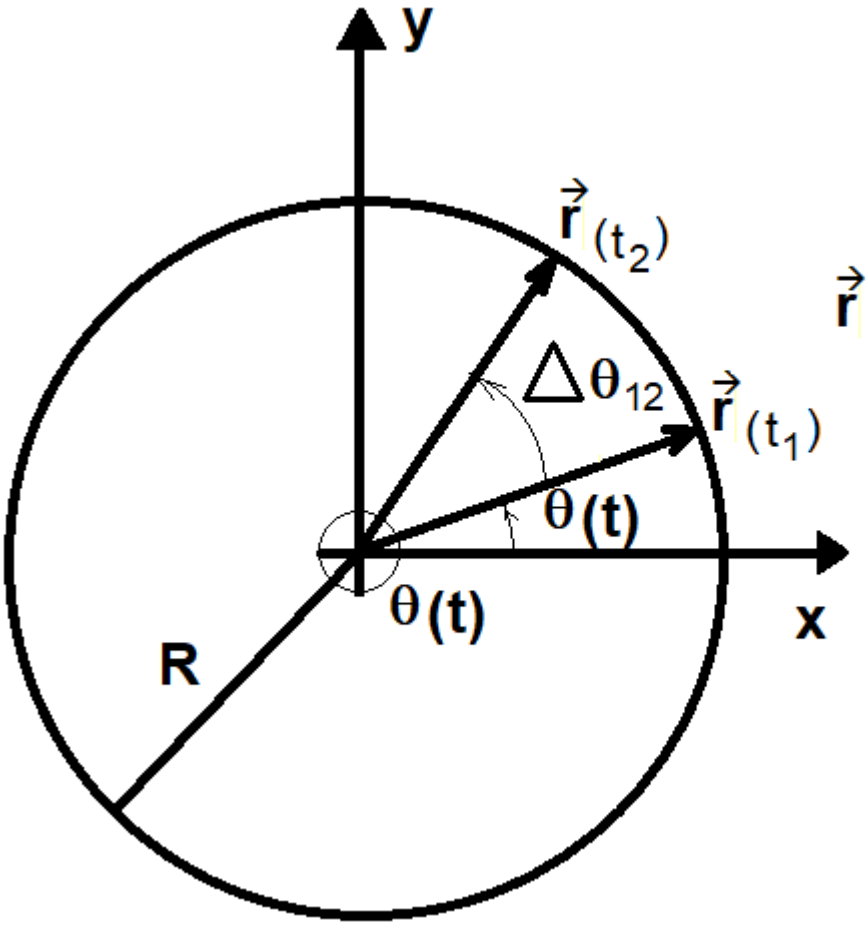


$\vec{r}\,(x;y)$

$\vec{r}\,(|\vec{r}|;\theta)$

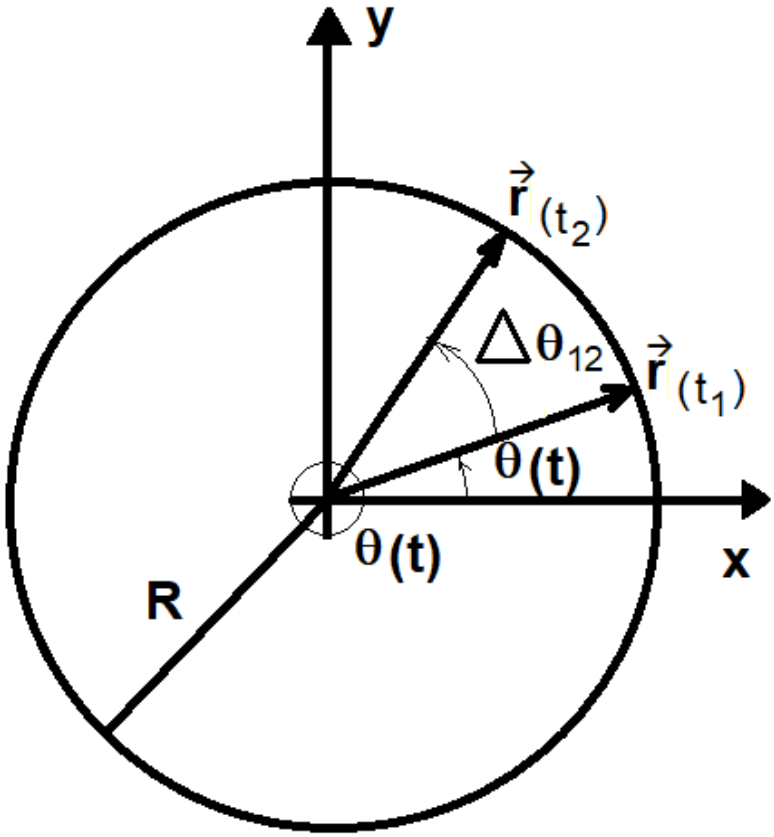
$\theta(t)$

$|\vec{r}| = R$



$\vec{\theta}(t)$

$\vec{r}(|\vec{r}|;\theta(t))$



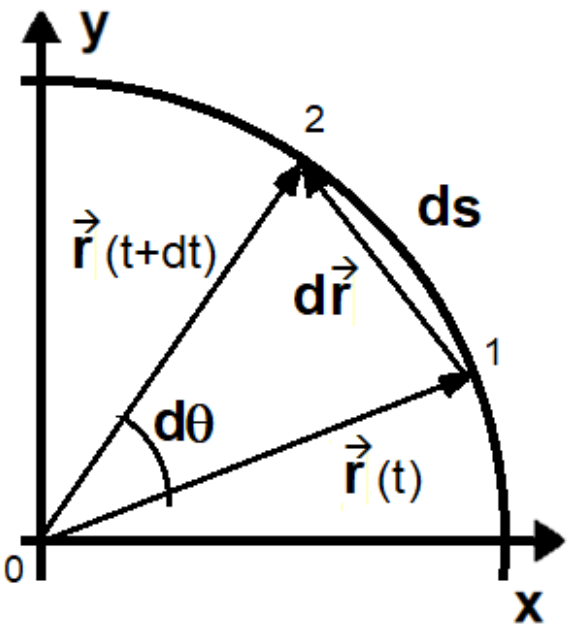
$$\vec{\theta}(t)$$

$$\vec{r}\,(|\vec{r}|;\theta_{(t)})$$

$$\vec{\omega}_m = \frac{\Delta\vec{\theta}}{\Delta t}$$

$$\vec{\omega} = \lim_{\Delta t \to 0} \frac{\Delta\vec{\theta}}{\Delta t} = \frac{d\vec{\theta}}{dt}$$

$$\vec{\alpha} = \lim_{\Delta t \to 0} \frac{\Delta\vec{\omega}}{\Delta t} = \frac{d\vec{\omega}}{dt}$$



$$|d\vec{r}| \simeq ds$$

$$|\vec{v}| = \frac{|d\vec{r}|}{dt} \simeq \frac{ds}{dt}$$

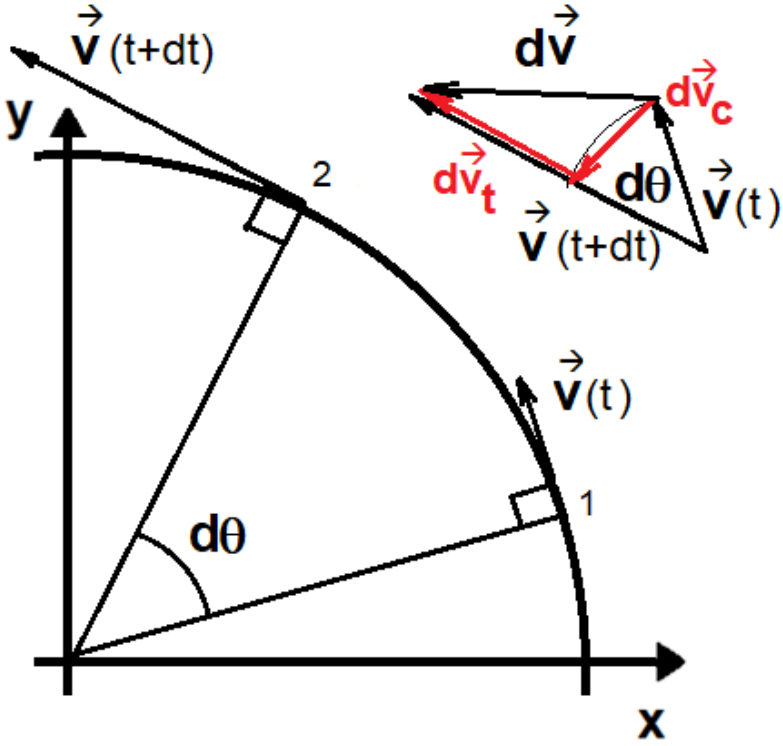
$$|d\vec{\theta}| = \frac{ds}{|\vec{r}|} \rightarrow \frac{|d\vec{\theta}|}{dt} = \frac{ds}{dt} \cdot \frac{1}{|\vec{r}|}$$

$$|\vec{\omega}| = \frac{|\vec{v}|}{R}$$

$$|\vec{r}| = R$$

$$\boxed{|\vec{v}| = |\vec{\omega}| \cdot R}$$

$$|\vec{v}(t+dt)| \neq |\vec{v}(t)|$$



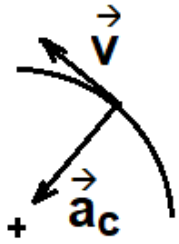
$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$d\vec{v} = d\vec{v}_c + d\vec{v}_t$$

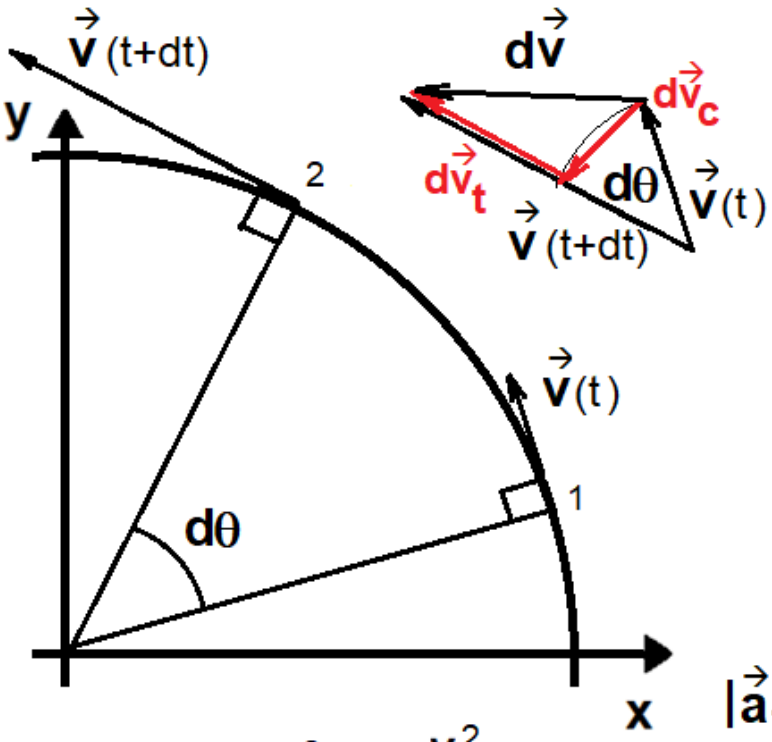
$$d\vec{v}_c = d\theta\,|\vec{v}(t)|$$

$$|\vec{a}_c| = \frac{|d\vec{v}_c|}{dt} = \frac{d\theta}{dt}\,|\vec{v}(t)|$$

$$|\vec{a}_c| = |\vec{\omega}|\,|\vec{v}(t)|$$



$$|\vec{v}(t+dt)| \neq |\vec{v}(t)|$$



$$a_c = \omega . v = \omega^2 . R = \frac{v^2}{R}$$

$$a_t = \alpha . R$$

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$d\vec{v} = d\vec{v}_c + d\vec{v}_t$$

$$|d\vec{v}_t| = |\vec{v}(t+dt)| - |\vec{v}(t)|$$

$$|d\vec{v}_t| = \omega(t+dt).R - \omega(t).R$$

$$|d\vec{v}_t| = (\omega(t+dt) - \omega(t))R$$

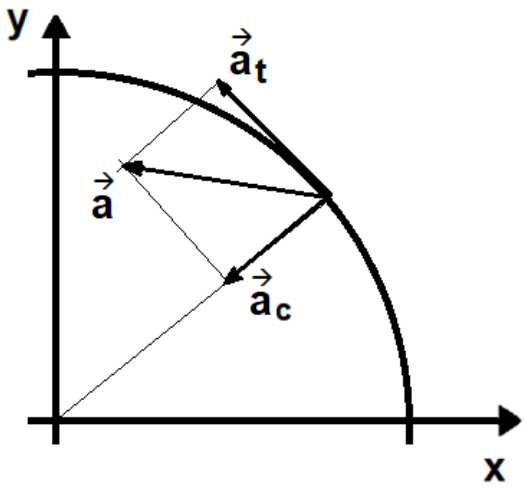
$$|\vec{a}_t| = \frac{|d\vec{v}_t|}{dt} = \frac{d|\vec{\omega}(t)|}{dt} R = |\vec{\alpha}| R$$

$$\boxed{|\vec{a}_t| = |\vec{\alpha}| R}$$

$$a_c = \omega . v = \omega^2 . R = \frac{v^2}{R}$$

$$a_t = \alpha . R$$

M.C.U. ==> $\alpha = 0$



$$\frac{d\vec{v}}{dt} = \frac{d\vec{v}_c}{dt} + \frac{d\vec{v}_t}{dt} = \vec{a}$$

$$\vec{a}_c + \vec{a}_t = \vec{a}$$